An automation shock in a small economy

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5 November 2026

Abstract

I propose to build a small, closed, agent-based economy in which a single real phenomenon — the *automation of production* — is the only internal shock, and to ask whether that one change is sufficient to produce the two movements Irving Fisher separated: a fall in the price of goods, and a debt-deflation in the price of assets. The economy is stock-flow consistent: every unit of money is somebody's asset and somebody else's liability, created only when a bank lends and destroyed only when a loan is repaid. Every agent is boundedly rational and sees only its own books; no agent forecasts the macro-economy. The claims of interest — deflation, structural unemployment, a debt-deflation spiral, and a grim mortality-driven floor beneath it — are things the model should be *able to produce from micro-rules*, never assumptions written into its equations. This document specifies the entities, the markets, the money, and the clock in enough detail to implement; every design question is resolved, and what remains is numeric calibration.

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1. The proposal

1.1 The question

The nightmare scenario is simple to state and hard to produce honestly. A generation of machines arrives that makes *more* goods with *fewer* workers. Supply rises and employment falls at the same instant; the newly unemployed stop spending, so demand falls while supply climbs. Prices fall — deflation. But debt is a fixed sum of money, and the interest on it must still be paid out of a money supply that is now shrinking. Repaying yesterday's loan in today's dearer money raises the *real* burden of every debt, which pushes households and firms to sell assets to raise cash, which deflates asset prices further: a debt-deflation spiral. High real rates kill the incentive to start or grow a business, investment collapses, and the economy settles into a depressed state.

The model's job is not to assert this story but to see whether it *emerges*. I will give agents local rules — buy food before anything else, lower your price when goods go unsold, refuse a loan that looks unsafe — and watch whether the macro-pattern above appears without anyone being told to produce it. The single internal cause is one declining parameter: the number of operators a machine of a given type and generation requires.

1.2 Modelling stance

Three commitments shape every later choice.

Agent-based. The unit of the model is the individual household or firm, acting on its own accounts, the prices posted so far, the calendar, and its own recent sales. There is no representative agent and no central auctioneer.

Stock-flow consistent (SFC). Money is not a free-floating price level. It is created when a bank writes a new deposit against collateral, and destroyed when that loan is repaid; a loan between two individuals only *moves* money that already exists. Every flow leaves an accounting trace in two places. A slump shrinks the money supply because banks make fewer new loans, not because

anyone's cash is annihilated.

Boundedly rational; the market prices risk, not the agent. No agent holds a model of the economy or forecasts the cycle. A bank is a mechanical secured lender that checks a collateral ratio. A private lender applies a local heuristic to a borrower's disclosed accounts (does current revenue cover the monthly payment with margin?). Risk is priced by the *market* — a borrower posts terms, lenders accept or decline, and a firm that cannot place its debt re-posts at a higher rate next turn — never by a forecast inside anyone's head.

2. The world

2.1 Entities and resources

The economy contains four kinds of agent: **individuals** (households, some of whom are also entrepreneurs or mine owners), one or two **banks**, and a **government**. Everything they trade falls into a small set of resource classes.

- **Consumer goods and services**, bought to raise welfare: staple food (survival), luxury food and private healthcare (general health), nightclubs and bigger housing (happiness), and private tuition (education).
- **Real estate** — houses of various sizes — which is both a place to live (a requirement, below) and an illiquid asset.
- **Liquid financial assets**: private debt, government debt, and bank shares. These trade every month.
- **Managed productive assets**: *mines* (mining land plus its finite stock of raw material) and *businesses* (a bundle of machines and the employees who operate them).
- **Machines**, which come in distinct *types* and, within a type, successive *generations* (levels). The generation is the automation dial of §3.2.
- **Raw materials** — a common and an advanced type — both consumed to build machines, the advanced one increasingly at higher generations (§4.9).

A household's welfare is three stocks — health, happiness, education — and its survival turns on a fourth necessity, food, bought before anything else. Individuals pay no tax; firms do.

2.2 The agent archetypes

The same individual can wear several hats; the archetypes below are roles, not castes, and the model lets agents move between them.

- **An ordinary individual** holds a job (or does not), lives in a house it owns or rents, carries welfare stocks, and each month must feed itself before it spends, invests, or saves.
- **An entrepreneur** is an individual who also owns a business outright — machines plus employees — and must meet the firm's obligations (wages, tax, debt service) before taking any profit home.
- **A tech entrepreneur** runs the special business that builds *machines* out of raw materials, employing engineers. This is the only agent that can introduce a new machine generation, so it is the supply valve of automation itself.
- **A mine owner** holds mining land, must post a selling price and quantity for its raw material each month, may sell the mine on the secondary market, and may deposit it into a bank in exchange for bank shares.
- **The indigenous population** are the initial mine owners: at the start they own all mining

land and the mines, which makes them the economy's first rentier-financier class (§7.2).

- **The government** taxes sales, employs public doctors, teachers and bureaucrats, runs the public hospital and university, borrows when short, and provides health and education in kind — but never cash, housing, or food.
- **The bank** is a mechanical secured lender whose capital base is deposited mining land, who may lend up to a multiple of that base, and who pays its depositor-shareholders a yearly dividend.

3. How a person lives

3.1 Welfare: health, happiness, education, and the hard zeros

Each lever of consumption does exactly one thing; there are no cross-effects to blur the accounting. The luxury supermarket and the private hospital raise *general health*; the nightclub and a bigger house raise *happiness*; teachers (public or private) raise *education*; the ordinary supermarket buys staple food, which is survival rather than welfare. Doctors are a special case: they do not raise general health at all — they *cure diseases* (§4.4).

Two couplings turn these tidy stocks into a trap. First, **happiness caps health**: there is a ceiling on how healthy an unhappy agent can be, so even a single-minded health-maximiser must buy some happiness to lift the ceiling before more medicine or luxury food does anything. Second, there are two **hard-zero** states that pin health at zero and hold it there — *homelessness* and an *uncured disease*. While pinned, no amount of luxury food helps; the condition must be resolved first (get housed; or buy the specialist cure). And because employers prefer healthier applicants (§4.5), an agent at health zero is also unhireable. Job loss then lowers income, which costs the agent its housing or its health, which makes it harder to be hired — a micro-level poverty spiral that is the household mirror of the macro debt-deflation spiral.

Homelessness itself does not kill; it sends health to zero but the agent lives, and public services still reach it, so the trap is escapable. What kills is a sustained inability to buy food, under a **three-turn rule**. An agent that cannot afford food does not die at once: it enters starvation, and the engine automatically force-sells its *liquid* assets — cash first, then bank shares, debt holdings, and directly-held mine land — at a below-market distress price to feed it. Only after three consecutive foodless turns, once nothing liquid remains to burn, does it die. Because houses can be sold only once a year (§6.2) and are not force-sold here, an asset-rich but cash-poor owner whose only wealth is an unsellable house still dies this way — death by *illiquidity*, not insolvency (§4.6). The same three-turn rule governs a newborn that cannot be fed (§6.1).

3.2 Careers, education levels, and the machine-operator bijection

Every productive job is the operation of a machine, and machine types stand in **bijection** with worker classes: a bureaucrat runs the admin machine, a doctor the medical machine, a teacher the teaching machine, and retail and hospitality staff theirs, while an *engineer* runs the meta-machine that builds machines. The thing that automates is itself a machine with operators who can, in turn, be automated. Mining is the exception: extraction is *labour-free*, a mine producing raw material directly from its land and finite stock with no operators at all (§4.9).

Alongside these productive workers, every business also employs **bureaucrats** who run its admin machine and *support* the productive staff in a fixed ratio — β bureaucrats per productive worker, about one to one at generation 0 (a hospital with twenty doctors runs twenty bureaucrats). Because the admin machine automates like any other, a higher generation needs fewer bureaucrats per unit of throughput, so β falls: the back office sheds workers exactly as the shop floor does.

Education is **per career**, and within a career it is a *level*. Your level in a career decides which *generation* of that career's machine you may operate: level 0 runs generation 0, higher levels run higher generations. An "advanced engineer" is therefore not a separate profession but simply a high-level engineer — the one who can build the next machine generation.

The friction that makes automation bite is that **switching careers resets education to zero**. Your level belongs to one career and does not transfer; a bureaucrat displaced by automation who retrains as an engineer starts again at level 0. This is what turns automation unemployment *structural* rather than merely cyclical: the displaced cannot glide sideways into the jobs that are growing. Which career an agent educates toward is its own decision, made by looking at last year's job offerings and training toward whatever was most in demand — adaptive, backward-looking expectations rather than a forecast.

3.3 An ordinary individual's turn

Each month, an individual who is neither an entrepreneur nor a mine owner acts in a strict, necessity-first order:

1. **Get paid.** Collect all income due: wages (if employed), the monthly repayment on any debt it holds, mine-land yield, and — once a year, as an upfront lump — rent from any property let out. The government also offers, this turn, a rationed chance at a free public level-up or disease cure (§4.3, §4.4).
2. **Buy food, or starve.** The first act of every turn. An agent that cannot buy food enters starvation; the engine force-sells its liquid assets below market to feed it, and death follows only after three consecutive foodless turns (§3.1).
3. **Buy welfare and education** with what remains — health, happiness, education, in any amount supply allows, on top of whatever the government provided free.
4. **Invest** with what is still left, in the monthly capital market: mine land and debt (§4.10). Houses are *not* bought here — they trade only on the annual real-estate day.
5. **Save** whatever is left as cash.

This necessity-first ordering is one of three parallel turn-orders — the household's here, the entrepreneur's in §5.7, the government's in §5.11 — all sharing one principle: in a downturn each agent is squeezed from the bottom up, and the order decides what gets cut first.

4. The markets

4.1 One protocol, many venues

Every market in the economy runs the *same* disequilibrium protocol, so no market ever clears instantaneously and gluts and shortages are both possible and informative. Each turn:

1. **Sellers post** their prices and quantities first.
2. **Buyers decide** what and how much to buy.
3. If demand exceeds supply, the good **sells out** and a **random lottery** rations who is served — rationing by chance, not by willingness to pay.
4. Sellers **reprice from last turn's feedback**: raise what sold out, cut what went unsold, and undercut or match what competitors charged.

One detail matters for the crash: the protocol must be **symmetric**. Just as oversubscribed demand raises prices, an oversupply of sellers with too few buyers must force asking prices *down*. This

falling-price branch is the asset-collapse channel; without it the debt-deflation spiral cannot run.

4.2 The consumer market (monthly)

Food, entertainment, private health, and private education all trade monthly under the common protocol: providers advertise, buyers choose, goods can sell out and trigger a lottery, and prices adapt. Housing is the one consumer good that does *not* trade here — it has its own annual day (§4.6, §4.7).

4.3 The education market: public and private

Education raises a citizen's level in one chosen career. Teachers are generic — a teacher can teach any subject — so what is produced is set by which subjects the government and universities choose to staff.

Public education is a policy lever. Each turn the government picks which teacher-types to employ, setting the mix of education produced. Citizens apply for the units needed to reach their next level; the government fills this demand in a priority order by housing status — **homeless first, then renters, then owners** — with a **lottery** inside each tier, and allocation is **all-or-nothing**: a winner completes a full level-up, a loser gets nothing. So free education is real but rationed, and only in the types the government chose to produce. *Private* tuition buys the specific career-type you want outright, escaping both the mix and the lottery.

Because a career switch resets education (§3.2), the government's *mix* choice shapes the future labour force: keep training bureaucrats when the economy needs engineers and you mis-train the population for obsolete work. Steering the mix toward growing careers is the main public tool for absorbing the automation shock.

4.4 The health market: disease, doctors, public and private

General health is bought at the luxury supermarket; *disease* is a separate, catastrophic state. There are five disease types, and each turn disease strikes citizens at random, sending the victim's health to a hard zero until cured. A cure requires buying ten units of the *matching* specialist doctor (five diseases to five specialties), at a public or private hospital.

Doctor supply flows from the education system — a generic teacher can train any specialty — and the government chooses its doctor-type mix each turn, just as it chooses its teacher mix. Free public healthcare then works identically to public education: apply, be served homeless-first with a lottery inside each tier, all-or-nothing (the full ten units and a cure, or nothing). Private healthcare buys the needed specialist outright. The poor therefore depend on the homeless-first lottery while the rich self-insure privately, and a government whose doctor mix mismatches the diseases actually circulating leaves some citizens curable only by paying.

4.5 The job market (annual, on the special day)

Hiring happens once a year, in the job step of the annual special day (§6.2) — after real-estate sales, before the rental market; between special days there is no hiring, though firms may shed workers in any month. The incumbent in a post has first preference and may keep it at the same salary without reapplying. An agent may also apply elsewhere, provided it meets the job's **minimum education** — the hard gate on who may apply (§3.2). Among non-incumbent applicants the **healthiest wins**: health is both a welfare stock and a hiring edge, which is why a hard-zero agent is unemployable. A worker raises its wage only by winning a higher-paid listing on job day, not by continuous bargaining. A laid-off worker cannot be rehired until the next annual job day, so

unemployment can last up to twelve months — making job loss a durable, dangerous event that savings must bridge. Because rent is paid a year upfront, a worker can lose its *job* mid-year but not its *house*: unemployment is a monthly risk, homelessness only an annual one.

4.6 The real-estate sale market (annual) and the illiquidity death

On the special day, real-estate **sales come early** — before the job market, since a sale needs no job (§6.2). Houses are bought and sold only on this day — when multiple buyers compete the winner is chosen at random — so real estate is illiquid between days. That illiquidity is precisely what enables the asset-rich, cash-poor **illiquidity death** of §3.1: an owner who runs out of cash in month three cannot reach its wealth until the special day months away, and houses are not among the liquid assets the engine force-sells to feed a starving agent. Mine land and debt, by contrast, trade monthly, so a house is the asset that can kill its owner by being unsellable. Once sales settle, each owner — the buyer who just acquired it, or the owner who kept it — **decides whether to occupy it or rent it out**; only properties offered for rent enter the rental market, which runs later, after the job step.

4.7 The rental market (annual, last on the special day)

Renting is settled last on the special day — after sales, the occupy-or-rent decision, and the job market (§6.2), so that applicants' salaries are known. The owner re-advertises the rent; the sitting tenant has first preference to renew (unless the home was sold out from under them); among other applicants the owner prefers the one with the **highest salary or job title**. Rent is paid **upfront for the whole year**: a tenant who cannot cover a full year does not secure the home, and an agent left with no dwelling has its health go to zero (homelessness). Because the year is paid in advance, housing is then secure until the next special day — there is no mid-year eviction. Ranking applicants by income means that in a downturn landlords swap unemployed tenants for higher-paid ones, so the weakest lose housing first.

4.8 The machine market (annual, on job day)

Tech companies sell machines to the other businesses, competing on price: several producers list prices, businesses place orders, and machines can sell out and push producers to raise prices next turn. Each machine carries both a purchase cost and a maintenance cost. Machines trade on *job day*, so a firm reconfigures its capital and its workforce in the same annual act — a machine purchase is inseparable from the hiring or firing of its operators, and hiring is annual.

4.9 The raw-materials market (monthly)

Mines sell raw material to tech companies every month under the common protocol. Extraction is **labour-free**: a mine employs no one and produces raw material directly from its land and finite stock. Each mine advertises a price and may sell any quantity up to its remaining stock — as much as it likes, at whatever price it posts; offering less than the stock is withholding, which props up the price and slows depletion. A buyer's order may carry a **roll-over** instruction: if it is not filled, the order persists and re-tries at a higher price up to a reservation ceiling, so a buyer that truly needs material bids itself up within its limit. Raw material is **finite** — each mine holds a stock sized to it, extraction depletes it, and the total across all mines is calibrated to be just enough to reach full automation. As mines deplete and post higher prices, input costs to the tech sector drift up: a slow long-run drag.

There are two raw-material types — a **common** input and an **advanced** one, one per mine type — and a machine's *bill of materials* depends on its generation: every machine needs the common input

in bulk plus some of the advanced one, and the advanced requirement rises with each generation. The advanced material is therefore a second automation bottleneck beside the engineers: as its mine depletes, each new generation costs more of a tightening resource — a material brake on automation that runs alongside the labour brake.

4.10 The capital and secondary market (monthly)

Three liquid assets trade monthly under the common protocol — **mine land**, **debt**, and **bank shares** — and for debt, primary issuance and secondary resale are the *same* market. There is only one debt instrument (§5.3); a holder reselling it lists a new price and the buyer, who sees the underlying borrower's current disclosure and the instrument's remaining payments, accepts or declines. No new claim is created on a resale; an existing claim changes hands. Because debt can always be resold, it carries a market price that tracks the borrower's health: as a borrower's accounts deteriorate, the resale value of its debt falls and holders mark it down and dump it *before* any default — a debt-repricing channel that feeds the asset collapse directly, not only through defaults. Participation on the lender side is open to any agent with enough savings; it is gated by a savings threshold, not by class.

5. Money, finance, and the firm

5.1 Where money comes from

The monetary spine is simple and strict. A **bank loan creates money**: the bank writes a new deposit from nothing against collateral. **Repayment destroys money**: the deposit is extinguished as the loan is paid down. A **private loan creates no money**: it only moves an existing deposit from lender to borrower. It follows that a boom, with banks lending freely, expands the money supply, and a bust contracts it — not because anyone's cash is burned, but because new lending dries up and existing loans are repaid faster than replaced. A default, by contrast, mainly destroys *bank equity* rather than money: the deposit the borrower already spent is still in circulation. This distinction — repayment shrinks money, default shrinks the bank — is what makes the deflation and the banking fragility two separate stories that happen to coincide in a crash.

5.2 Mortgages and bank loans

To borrow from a bank you must pledge **collateral**: real estate or mining land only, never machines.[1] The bank advances a lump sum and is repaid in an equal monthly stream to a deadline, the total exceeding the lump sum by the interest. The bank is a **mechanical** lender: it lends up to a loan-to-value fraction of the collateral, marks the collateral to market each turn, and issues a margin call when the collateral falls below the loan. The borrower meets the call by **selling liquid assets first** to pay the loan down; the bank forecloses only if those cannot cover the shortfall. It runs no cash-flow forecast and holds no model of the economy; the loan-to-value haircut is its entire risk model. Pledged collateral carries a **lien** that blocks sale — you keep the use and the income of a liened house or mine, but you cannot sell it until the loan clears, which sharpens the illiquidity risk for an entrepreneur who pledged a home to start a firm.

5.3 Private debt

The direct debt market is where forward-looking credit lives, though even here no one forecasts the macro-economy. An issuer exposes all its accounts as a structured disclosure; it then offers

[1] Machines are wrong-way collateral: a firm usually fails *because* its machines are an obsolete generation — worth least exactly when seized — so the bank will not lend against them.

an instrument — a lump sum now, a stream of equal monthly payments to a deadline, the total exceeding the lump sum by the interest — and a lender reads the disclosure and the offer and either accepts or declines, with no negotiation. If no lender bites, the issuer re-posts next turn at a higher rate, so a distressed firm visibly ratchets up its offered rate to place its debt. Direct-market loans are unsecured and transferable: the lender bears the loss if the borrower fails, which is why this debt must carry a risk premium over collateralised bank lending.

	Bank loan	**Private loan**
Security	secured (collateral)	unsecured
Credit test	mechanical LTV gate	lender's heuristic on disclosure
Transferable	no (held to maturity)	yes (resold on the capital market)
Effect on money	creates money	moves existing money
Loss falls on	the bank's equity	the individual lender

Table 1: The two credit instruments.

5.4 Government debt

A government bond is not a special instrument; it is exactly the private-debt instrument above, issued by the state. The government exposes its tax revenue and its salary obligations, offers a lump sum, rate, and monthly schedule, and savers accept or decline. Only the disclosure differs. This is what makes the fiscal rescue of §5.11 endogenous: whether the state can borrow through a downturn is a verdict of the same market that prices every other debt.

5.5 Becoming an entrepreneur

Entrepreneurs are not a fixed caste; an ordinary citizen can enter. The gate is **property**: you must own real estate, because the canonical way to raise start-up capital is to pledge that real estate to a bank and take a loan. Renters, owning no collateral, cannot enter. Entry is therefore debt-funded — new firms start leveraged — and the stake is the owner's home: because the start-up loan is secured on it, a failure (§5.8) seizes the *pledged* property, though any other, unmortgaged property is shielded — and the owner keeps a food reserve and recovers at the next real-estate day by mortgaging or selling that home. Once running, the firm behaves like any business: it buys machines, pays wages and taxes, and earns by selling its products. Entry makes the number of firms endogenous, and it is the recovery channel after a crash: those who kept their property through the bust can pledge it and start again.

5.6 The tech entrepreneur's decisions

The tech entrepreneur is the supply valve of automation, facing three nested choices each period.

- **Product strategy.** *Stay* (same machines, same volume), *expand* (more of the same current-generation machines, hiring more regular engineers — the only growth path before automation ignites, and the one that *adds* jobs), or *upgrade* (hire a scarce advanced engineer to raise one product to a higher, pricier generation — the sole way a new generation enters the economy).
- **Process and reinvestment.** When it upgrades its *own* production machines, which lowers the operators each needs, it either *upgrades and fires* (same output, fewer engineers, bank the saving) or *upgrades and redeploys* (keep the engineers, make more product). Which it does turns on **expected demand**: a firm scales up only if it believes it can sell the extra output, and in a weak, deflationary state it sheds labour instead — which weakens demand further. The displacement-versus-expansion choice is endogenous, not assumed.

- **Personal portfolio.** Being also a household, it may instead take profit home — into happiness, health, mine land (backward integration into its own input supply), rental property, or others' debt.

5.7 The entrepreneur's turn

Each turn an entrepreneur acts in order: pay the per-unit sales tax on what the firm sold; service the firm's debt; decide whether to borrow (pledging real estate or mine land, never machines); and, on job day, decide whether to buy or upgrade machines and hire or fire operators. Only what remains after the firm's obligations and investment may go to personal saving, outside investment, or welfare. The ordering is the point: firm obligations precede firm investment, which precedes personal use, so a firm in distress is cut from the bottom up and only a genuinely insolvent one fails to meet its wages and debt.

5.8 Firm fold and the bankruptcy waterfall

When an entrepreneur can no longer meet its obligations, the firm folds. Only its *liquid* estate is marshalled — cash savings, debt receivables, bank shares, and directly-held mine land, all liquidated at distress prices — together with any specifically *mortgaged* collateral the bank has a lien on. **Unmortgaged real estate is not seized.** The proceeds meet claims in a strict priority waterfall:

1. **Wages first.** Labour is the senior claim; if the estate cannot pay all workers, random workers are paid until it runs out.
2. **Tax next.** The per-unit sales tax owed on units already sold is paid to the government *before* the bank — a defaulting firm still owes the state its tax.
3. **The bank**, recovering from the remaining liquid estate and from the mortgaged collateral it holds (sold slightly below market, pushing house prices down) and seized receivables (sold on the debt market). It does not reach unmortgaged property.
4. **Other creditors** — unsecured direct-market lenders — take whatever liquid estate remains, usually nothing; the shortfall is their loss, money destroyed, which is why their lending carried a premium.

Unmortgaged property is shielded, and a food reserve is kept: a small survival reserve of cash — enough to buy food until the next real-estate day — is carved out before the waterfall, so a fold cannot immediately starve the owner. At that real-estate day the ex-entrepreneur recovers by either mortgaging its unmortgaged home or selling it, turning the shielded illiquid asset into cash. So a folded owner with an unencumbered home carries its reserve to the next real-estate day and recovers; it dies the three-turn starvation death only if it has neither liquid assets nor unmortgaged property. Seizing receivables *before* default closes a laundering loophole (borrow, re-lend, fold, keep the receivable nets to zero) and is the reason the model needs a secondary debt market at all: the bank must be able to liquidate what it seizes.

5.9 The mine owner's turn

A mine owner must, each month, post a selling price and a quantity of raw material (at most its remaining stock) and sell to tech companies under the common protocol. It may also sell the mine itself on the monthly secondary market, or **deposit** it into a bank. Depositing converts the mine into **bank shares**: the physical land becomes the bank's permanent capital and can never be withdrawn, but the depositor receives a claim on the bank's yearly dividend, which it can sell on the monthly share market for liquidity. Whether to keep the mine (liquid, pledgeable, sellable for emergency food money) or deposit it (a higher but riskier, locked-up, once-a-year return) is the

mine owner's central financial choice (§5.10).

5.10 The bank's structure and its scripted decisions

The bank's capital base is **mining land**, not cash: any agent may deposit a mine, the bank operates the deposited mines and sells their raw material, and it may lend up to **ten times** the value of its holdings, creating money as it lends. Depositors are the bank's shareholders; the bank pools its income — raw-material sales plus loan interest, minus default losses — and pays it out as a yearly dividend pro rata, on the special day (§6.2). Because shares trade *before* the dividend that day, a share sold then carries the dividend to its buyer. This is the one place a dividend exists, companies paying none, and it is a strong reason to deposit.

The bank is designed to be **run-proof from the depositor side** but not from the asset side. No depositor can withdraw capital; for liquidity it sells its share instead, so there is no classic run. But the bank's equity still absorbs loan losses, and because seized collateral (real estate) is illiquid, the bank raises cash by *selling its own mine-land capital* on the monthly market. In a bust this becomes a **fire-sale amplifier**: losses force the bank to dump mine land, mine-land prices fall, the value of its remaining base falls, the ten-times ceiling falls, and lending contracts further — a mine-land debt-deflation that runs alongside the one in houses. The crash bites the shareholders through a collapsing dividend, a falling share price, and a shrinking base at once; cumulative losses that exhaust equity freeze credit entirely.

Two design hazards are worth naming because they constrain the implementation. First, a **self-valuation loophole**: if lending is sized off the posted price of *unsold* reserves, and if the deposit incentive drove everyone to deposit so the bank *is* the raw-materials market, the bank could post a high price to inflate its own lending base and mint money against collateral it values itself. The rule that avoids this is to **mark the deposited mines to the independent mine-land market**: value each at its remaining stock times a per-unit reference price drawn from recent *independent* mine-land sales of that type (adverse selection keeps the best mines independent, so that reference market stays liquid and outside the bank's control). The bank posts raw-material prices, never mine-land prices, so it cannot value its own collateral; the mark reuses the same mark-to-market the bank already applies to per-loan collateral, embeds each mine's remaining reserves automatically, and falls in a bust as mine-land prices fall — feeding the amplifier above. (An optional floor at trailing realised revenue can be switched on if a mine ever marks high while selling nothing.) Second, the reason **not everyone deposits**: the monthly cash of independent sales beats a yearly dividend for a household that must eat every month; deposited capital is locked and cannot be pledged or sold for emergency food; pro-rata pooling dilutes the best mines; and the dividend turns safe mine income into a risky profit-share that craters in a crash. These disincentives bite hardest on exactly the agents who would otherwise cause a monopoly — the best mines and the cash-poor — so the equilibrium is a partial, adversely-selected deposit base, which both prevents a raw-materials monopoly and preserves a competitive price to mark the lending base against.

5.11 The government's scripted decisions

The government is scripted but not omnipotent. Its income is the per-unit sales tax plus borrowing; its spending is the wages of public doctors, teachers, and bureaucrats, plus interest on its debt. There is **no cash welfare** — but public health and public education are universal in kind and reach even the homeless, and are therefore the poverty trap's circuit-breakers. Each turn it acts in order: receive tax; **pay its workers or fire them** (firing is its adjustment valve, and public-sector layoffs feed straight into unemployment); pay debt interest; issue new debt to cover any shortfall;

and distribute the health and education it produced.

Its role in the cycle is an **endogenous** stabiliser-or-amplifier. It would rather borrow than fire, but whether it *can* is decided by the debt market: a saver reading falling tax revenue against high salary obligations may decline or demand a punishing rate. If savers lend, the state maintains employment but now owes interest to the wealthy — a concentration channel; if no one lends tolerably, it is forced into austerity — the amplifier outcome. The timing is cruel, because the same downturn makes the state want to borrow, worsens its creditworthiness, and makes savers most reluctant (in deflation, holding cash earns a positive real return). And because it too owns machines and can automate, the public sector sheds its own workers into a system with no welfare to catch them. The free health and education each citizen receives is public production divided by population, so austerity, too few public machines, or a growing population all shrink it — weakening the circuit-breaker exactly when it is needed most.

6. Population and time

6.1 Birth, death, and the dark stabiliser

Fertility is **propertied**: only owners may have children, renters cannot, so over generations the population descends increasingly from owners. A newborn is endowed at birth with a fixed **required gift** — ten units of real estate (its own dwelling) plus a required cash gift G — and starts as a whole new household in the lowest job tier, housed and itself fertile once it clears the gate below. To have a child, a parent must hold, over and above its own housing, at least **ten spare units** of real estate *and* at least $3G$ **in liquid assets**, with another ten units and $3G$ for each further child; whether an eligible parent in this voluntary band actually reproduces is governed by its optimization strategy. Above a ceiling — **one hundred units of real estate and** $10G$ **in liquid assets** — reproduction is *mandatory*: the very rich are forced to have a child, a strong anti-concentration bleed on the top. On birth the engine transfers the land and raises the cash gift from the parent's liquid assets, force-selling them at aggressively falling prices until the gift is funded; and a newborn that cannot be fed is subject to the same three-turn starvation rule as any agent. Real-estate accumulation and household formation are the same process.

Death routes an estate in one of two sharply different ways. With an heir, the firstborn inherits everything undivided — **primogeniture**, a concentrating force. With *no* heir, the estate splits three ways: real estate is broken into ten-unit pieces and handed to random homeless people (instantly housing and enfranchising them); **all cash and liquid assets escheat to the government** as revenue — including the receivables, which the state then collects; and the decedent's own debts are **written off**, so its creditors lose. A childless death is thus strongly anti-concentration, the opposite of primogeniture, and the distributional fate of the whole model hinges on the mix of the two.

This yields the model's most distinctive emergent claim. Because a childless death writes off the decedent's debts and de-levers the top, a **death-wave in a crash cancels debt en masse** — a grim jubilee by mortality. The deeper the slump, the more deaths, the more compounding debt simply vanishes, easing the burden on survivors and handing houses to the bottom. A second, fiscal stabiliser rides alongside: the liquid wealth of the childless dead refills the treasury, so a death-wave strengthens the public safety net exactly when demand for it peaks. The spiral of §3.3 is therefore not purely divergent: mortality is a genuine negative feedback, a Malthusian damper, and its losers are disproportionately the lenders whose loan books the deaths destroy. The model should be allowed to reach this floor honestly, through attrition, and must not be rescued by any

scripted recovery.

Hypothesis. The economy can find a floor beneath a debt-deflation without fiscal rescue, through mortality that de-levers it — a smaller, poorer, de-levered population, not a clean recovery.

6.2 Time and scheduling

The clock is twelve months to a year, one turn per month. Most activity — food, welfare and education, the government's distribution, mine-land and debt trading — happens every month. But once a year a single **special day** gathers the slow markets into one strict sequence, each step feeding the next: (1) **bank shares** trade, before the dividend, so a share sold now carries this year's dividend to its buyer; (2) the **dividend** is paid to the post-trade holders; (3) **real estate is bought and sold**, after which each owner chooses to occupy or let; (4) the **job market** opens (hiring and machines); and (5) the **rental market** matches tenancies — last, because applicants are ranked by the salary of the job they just did or did not secure. House *sales* precede the job market (a sale needs no job), while *renting* follows it. This annual rhythm has three consequences the model should exhibit:

- An **annual liquidity cliff**: a full year's rent falls due upfront on one day, so a worker who lost income needs a whole year's buffer to stay housed, and real-estate day becomes a synchronised reckoning where the cash-poor are evicted en masse — a sharp annual eviction cliff rather than a smooth decline.
- **Tenant-sorting by income**: because landlords rank applicants by salary, a downturn lets them swap weak tenants for strong, so the weakest lose housing first.
- **Frequency separation**: housing adjusts annually while labour, goods, and capital adjust monthly, so housing is the slow variable, and the whole cycle may phase-lock to this annual forcing, with crashes clustering at year boundaries.

7. Behaviour, the start, and the experiment

7.1 Optimization strategies and the social classes

Each agent carries one objective, drawn at random at birth and not inherited, and the dependency chains mean none is truly single-minded. A *happiness*-maximiser buys bigger houses (withdrawing them from the rental pool) and nightclubs; a *health*-maximiser must also buy happiness, because happiness caps health; an *education*-maximiser must also keep health up, because hiring is healthiest-wins — a three-link chain of education needing health needing happiness; a *savings*-maximiser hoards cash; an *investment*-maximiser buys income assets — rental property, mine land, debt. Note the opposite housing effects: happiness-maximisers withdraw real estate from the rental market while investment-maximisers supply it, so the strategy mix sets rental supply.

These objectives, crossed with housing and employment status, give a ladder of **social classes** whose fates the model is built to compare: homeless and unemployed (health zero, unhireable); renting and unemployed (one real-estate day from homelessness); renting and employed (housed but assetless, one shock from the trap); owning and employed (secure, accumulating); owning and unemployed (asset-rich, perhaps cash-poor, exposed to the illiquidity death); the **rentier** who lives off rents and liquid assets without working; the **entrepreneur** who has risked a home to run a firm; and the **indigenous** mine owner at the top. The through-line is **why you need savings**: with annual hiring, annual rent, and monthly food, only a cash buffer carries an agent across a gap in income — and running it down is the common path into the trap.

7.2 The initial seed

The economy begins stable, fully employed, and at level 0 — automation has not started and must be ignited from within. Every individual starts housed (owning or renting), with a job, with health and happiness at ten, and with the starting education its job requires. Wealth is drawn at random across three strata — a *bourgeoisie* owning their home plus extra real estate and debt holdings, a *middle* owning only their home, and a *poor* owning nothing and so renting — but this is asset ownership, not shelter: nobody is homeless at the start. The **indigenous** own all mining land and the mines, which makes them the economy's primary source of capital from turn one; they capitalise the bank, collect its dividend, and lend to entrepreneurs, so financial power is triply concentrated in them at the outset. Critically, there are regular engineers but **no advanced engineers**: that class starts empty, which is what forces ignition.

7.3 The automation shock, stepped through

With only level-0 machines and no advanced engineers, the only thing a business can do with profit at first is buy more level-0 machines — so the economy can only *expand*, hiring more operators for more output. It cannot yet automate. Ignition is endogenous and runs in sequence: an agent acquires enough education to become the first **advanced engineer**; a tech company hires it and builds a level-1 machine; businesses buy the new generation, which makes the same or more output with fewer operators; and the fire-or-redeploy choice finally has something to bite on. The shock's *timing* is therefore set by how fast the education system produces that first advanced engineer — so university funding and mix are the master lever on *when* the shock fires, not merely on how unequal its aftermath is.

From there the model should, if the micro-rules suffice, generate the whole arc without being told to: rising supply and falling employment together; unsold goods and falling prices; a shrinking money supply as lending dries up; a rising real burden of fixed debts; distressed firms ratcheting up offered rates and holders dumping marked-down debt; house and mine-land fire-sales as banks and folding firms raise cash; a bank base contracting through its own fire-sale; and a government whose rescue succeeds or fails at the verdict of the debt market. And beneath it all, the dark stabiliser of §6.1: a death-wave that cancels debt and hands houses to the bottom, letting the system find a floor through attrition.

Hypothesis. A single declining parameter — operators per machine — is sufficient, under boundedly-rational micro-rules and stock-flow-consistent money, to produce both of Fisher's movements: a deflation in goods and a debt-deflation in assets, with structural unemployment between them. The experiment is to run the economy with this change permitted and again with it forbidden, and compare posted prices, new lending, and deaths.

7.4 Resolved questions, and what remains to calibrate

The design is closed; every structural question has a decision. Resolved earlier: mining is labour-free with no miner career; the machine market trades on job day; mine-land yield is monthly, unbounded up to a mine's finite stock; rent is annual and paid upfront, so housing is secure within the year (no mid-year homelessness); fertility gates are fixed and the voluntary choice follows the agent's objective; the lookahead horizon follows an agent's time-preference; the bank's lending base is marked to the independent mine-land market (§5.10); agents choose their education from last year's job offerings; a margin call is met by selling liquid assets before any foreclosure; and the glut branch is just the protocol's "cut the price of what went unsold," so the symmetric price-fall is

already built in. Resolved here:

- **Secured-first in a childless estate**: the bank seizes its lien'd collateral first; only unsecured owed debts (and any deficiency) are written off, so the homeless receive houses net of any lien and the bank bears a loss only when collateral is worth less than the loan. A sub-ten-unit remainder escheats to the government with the liquid assets; an heir inherits the net estate only, never liabilities.
- **No forced deleveraging**: the ten-times ceiling governs only *new* lending (it stops as the base shrinks); existing loans are disciplined solely by per-loan margin calls against their own collateral — the bank never recalls the book en masse.
- **Labour-force inflow and retraining**: births are the only inflow (no immigration); a career switch resets education to zero and re-education then proceeds at the ordinary rate, so there is no separate retraining speed.
- **Scripted-agent objectives are deliberately the implementer's choice** — what the government optimises and an entrepreneur's reinvest-versus-extract split are left open by design, not by omission.

What remains is *numeric calibration*, not design: the required cash gift G (a newborn's endowment, default about a year of subsistence food), the bureaucrat support ratio β (default one at generation 0), the raw-material recipe coefficients per generation, and the usual constants — the loan-to-value haircut, the ten-times leverage cap, the sales-tax rate, the fire-sale discount, the total raw-material stock, and the t_0 counts. Each is a knob to set and sweep, with no design decision outstanding.